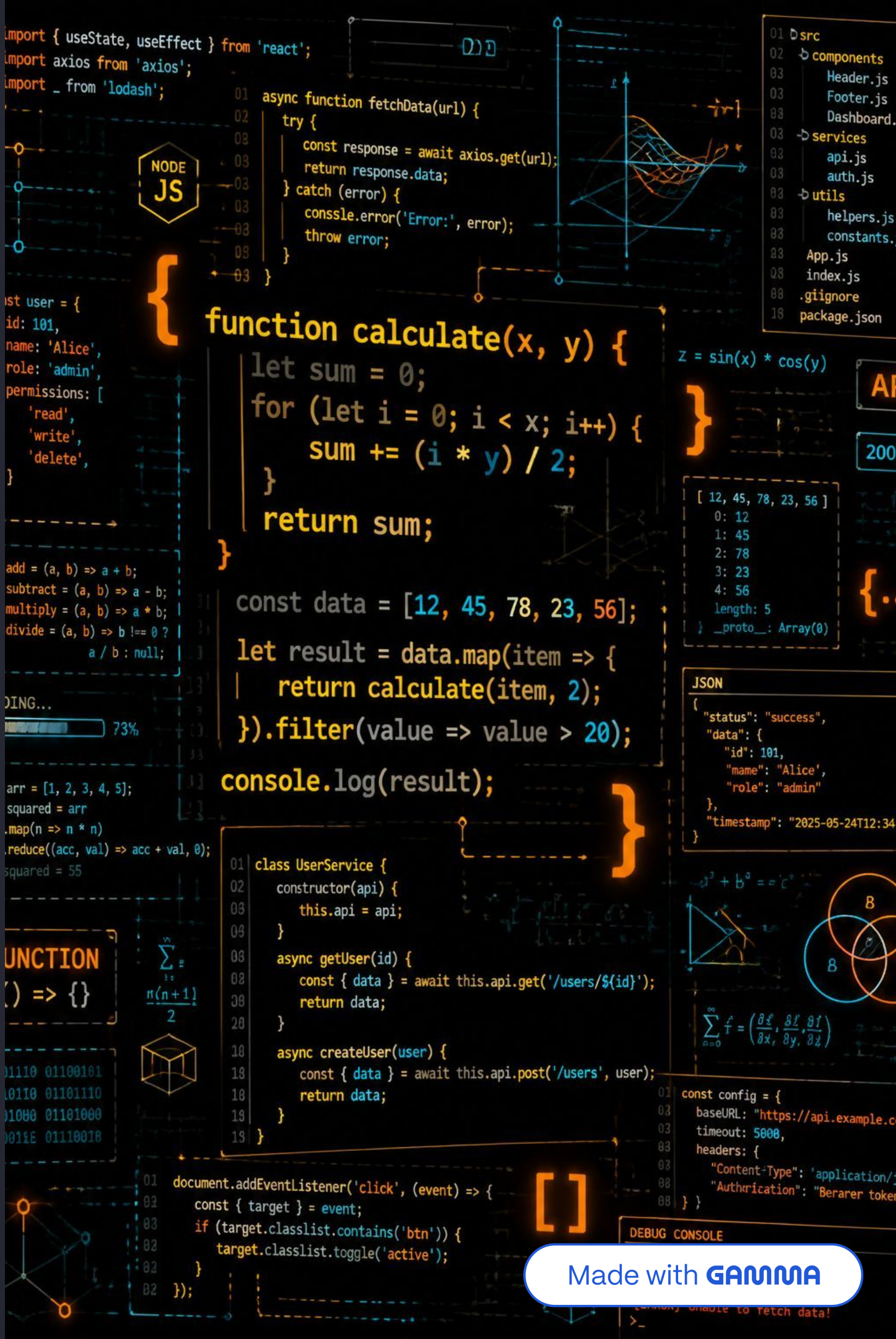
JavaScript Chapters 27–30

Random Numbers & Number Conversions

Beginner Friendly Notes — Roman Urdu me, bohot aasaan aur simple tareeqay se

{ JAVASCRIPT

BEGINNER LEVEL



CHAPTER 27

Random Numbers

Random number woh number hai jo computer khud choose karta hai — har baar alag! JavaScript isay games, OTP, passwords aur lucky draws ke liye use karti hai.

Dice Game

1 se 6 tak random number

Lucky Draw

Winner select karna

OTP / Password

Secure code generate karna

Random Quiz

Sawaal randomly dikhana

CHAPTER 27 — MATH.RANDOM()

Math.random() Kya Karta Hai?

Code Example

```
let num = Math.random();
console.log(num);
```

Output: **0.673245** (har baar alag!)

Samjhein Step-by-Step

- **Math.random()** ek built-in function hai
- Output hamesha **0 se 1** ke darmiyan hota hai
- Har baar run karne par **naya number** milta hai
- Output kabhi 0.0 se kam ya 1.0 se zyada nahi hota

CHAPTER 27 — MATH.FLOOR() + DICE

1 Se 10 Tak Random Number

Code

```
let num = Math.floor(
  Math.random() * 10
) + 1;
console.log(num);
```

Output: **7** (1 se 10 ke darmiyan)

Formula Samjhein

- **Math.random() * 10** → 0 se 10 tak decimal
- **Math.floor()** → decimal hata kar poora number deta hai
- **+ 1** → range 1 se 10 ho jati hai

🎲 Dice Example (1–6)

```
let dice = Math.floor(
  Math.random() * 6
) + 1;
console.log(dice);
```

Output: **4**

CHAPTER 28

String Ko Number Mein Badlein

Kabhi kabhi number **text (string)** ki shakal mein aate hain — jaise "25". Unhein asli number banana zaroori hota hai taake math kaam kare!

parseInt() — Poora Number

```
let age = "25";
console.log(parseInt(age));
```

Output: **25**

parseFloat() — Decimal Number

```
let price = "25.75";
console.log(parseFloat(price));
```

Output: **25.75**

Extra Text? Koi Masla Nahi!

```
let px = "100px";
console.log(parseInt(px));
```

Output: **100** — px ignore ho gaya!

CHAPTER 29

Number() aur String()

Number() — String → Number

```
let num = Number("100");
console.log(num);
```

Output: **100**

String ko number mein badalta hai — math ke liye useful!

String() — Number → String

```
let age = 20;
console.log(String(age));
```

Output: **"20"**

Number ko text mein badalta hai — display ke liye useful!

typeof — Data Type Check Karein

```
console.log(typeof "20");  // string
console.log(typeof 20);    // number
```

Output: **string** aur **number**

CHAPTER 30

toFixed() — Decimal Control Karein

Kya Hai toFixed()?

Decimal ke baad kitne numbers dikhane hain — yeh control karta hai. Money, percentage aur marks ke liye bohot useful!

```
let price = 25.6789;
console.log(price.toFixed(2));
```

Output: **"25.68"**

Different Examples

toFixed(1)

25.6789 → **"25.7"**

toFixed(2)

25.6789 → **"25.68"**

toFixed(3)

25.6789 → **"25.679"**

⚠ ⚠️ toFixed() **string** return karta hai!

QUICK COMPARISON

toFixed() Comparison Table

Ek hi number par different `toFixed()` values ka asar dekhein:

Original Number	toFixed(1)	toFixed(2)	toFixed(3)
25.6789	"25.7"	"25.68"	"25.679"
9.999	"10.0"	"10.00"	"9.999"
3.14159	"3.1"	"3.14"	"3.142"

💡 Yaad rakhein: toFixed() ka output hamesha **string** hota hai — chahe number jaisa dikhe!

REVISION

Quick Revision Table

Saaray functions ek nazar mein:

Function	Maqsood (Purpose)	Example
`Math.random()`	0–1 random decimal	0.673245
`Math.floor()`	Decimal hata kar poora number	Math.floor(4.9) → 4
`parseInt()`	String → integer	parseInt("25px") → 25
`parseFloat()`	String → decimal	parseFloat("25.75") → 25.75
`Number()`	String → number	Number("100") → 100
`String()`	Number → string	String(20) → "20"
`typeof`	Data type check karna	typeof 20 → "number"
`toFixed()`	Decimal places control	(25.678).toFixed(2) → "25.68"

Common Mistakes & Thank You 🚀

→ **parseInt vs parseFloat**

Decimal chahiye to `parseFloat()` use karein, warna `parseInt()`

→ **toFixed() returns string**

Iska output number nahi, **string** hota hai — yaad rakhein!

→ **Math.random() har baar alag**

Har run par naya number milega — yeh normal hai!

→ **Number() vs parseInt()**

`Number()` poori string convert karta hai, `parseInt()` sirf pehla number leta hai

✅ Keep Learning JavaScript! Practice karte rahein — aap zaroor seekh jayenge 🎯